



Implementation of DHCP, DNS, Active Directory, and Group Policy for Network Management

Group : 7

Names : Aldi Syarif Alhakim
Alezunga M. Nadhif Pohan
Muhamad Ihsad Rasyad

Class : 2CS1

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PROJECT ON

Windows Server for Network Management

Developed by

- 1. Aldi Syarif Alhakim**
- 2. Alezuna M. Nadhif Pohan**
- 3. Muhamad Ihsan Rasyad**

Windows Server for Network Management

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Name of Faculty : Tri Agus Riyadi, S.Kom., M.T.

Names of Developer :

1. Aldi Syarif Alhakim
2. Alezuna M. Nadhif Pohan
3. Muhamad Ihsan Rasyad

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CERTIFICATE OF ORIGINALITY

This is to certify that the project report titled "Implementation of DHCP, DNS, Active Directory, and Group Policy for Network Management" is an original work completed by Aldi Syarif Alhakim, Alezuna M. Nadhif Pohan and Muhamad Ihsan Rasyad. This project has been submitted in partial fulfillment of their course requirement at the National Institute of Information Technology (NIIT). The project report has been prepared under our guidance and supervision, and it is ensured that the work presented in this report is the result of the individual efforts of the aforementioned students. The contents of this report have not been submitted to any other institution or organization for the award of any degree, diploma, or other similar recognition. Author acknowledge that the ideas, designs, and implementations presented in this project report are the intellectual properties of the students mentioned above. Any use or reproduction of this work must give proper credit to the original authors. Author hereby endorse the authenticity and originality of the work presented in this project report and confirm that it meets the academic standards and requirements set forth by the National Institute of Information Technology (NIIT).

Coordinator:

Tri Agus Riyadi, S.Kom., M.T.

BACKGROUND

In the modern era of information technology, the management of computer networks within an organization has become increasingly complex and critical. Organizations, whether educational institutions, companies, or government agencies, require a reliable, centralized, and efficient network infrastructure to support their daily operational activities.

Windows Server, developed by Microsoft, is one of the most widely used server operating systems in enterprise environments. It provides a comprehensive set of services that enable network administrators to manage users, devices, and resources in a structured and scalable manner. Among the most essential services offered are Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Active Directory Domain Services (AD DS), and Group Policy Objects (GPO).

DHCP plays a fundamental role in automating the assignment of IP addresses to client devices within a network, eliminating the need for manual configuration and reducing the risk of address conflicts. DNS, on the other hand, is responsible for translating human-readable domain names into IP addresses, enabling seamless communication between devices across the network.

Active Directory Domain Services (AD DS) serves as the backbone of centralized identity and access management, allowing administrators to create and manage user accounts, computers, and organizational units within a domain environment. Building upon AD DS, Group Policy Objects (GPO) provide a powerful mechanism for enforcing security settings, software configurations, and user restrictions across all machines and users within the domain.

This paper aims to explore and demonstrate the configuration of these four essential services — DHCP, DNS, AD DS, and GPO — within a Windows Server environment, providing a practical understanding of how they work together to build a functional and secure network infrastructure.

SYSTEM ANALYSIS

This paper presents the configuration of an Active Directory server environment utilizing core Windows Server services. The authors simulate the role of system administrators to build a fully functional demo environment leveraging Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Active Directory Domain Services (AD DS), and Group Policy Objects (GPO).

The entire environment is built and simulated using VMware Workstation as the virtualization platform, which allows the authors to run multiple virtual machines simultaneously on a single physical host. Two virtual machines are deployed within VMware — one acting as the Windows Server 2022 Domain Controller and the other as the Windows Server 2022 client machine. Both virtual machines are configured using Internal Network mode within VMware to ensure isolated and stable communication within the virtual environment, preventing any interference from external networks.

IP address assignment is automated through DHCP, while DNS handles domain name resolution to enable seamless connectivity between server and client. User accounts and domain resources are managed centrally through AD DS, where clients must authenticate using registered domain accounts. Security and access restrictions are enforced through GPO, which limits client users from accessing certain system features and settings.

The overall objective is to construct a fully operational demo environment within VMware Workstation that simulates real-world Active Directory server administration using essential Windows Server services.

Services

1. **Active Directory Domain Services (AD DS)**

A Microsoft service that centralizes user, computer, and network management in a domain environment, allowing administrators to manage authentication, security, and access control efficiently.

2. **Domain Name System (DNS)**

DNS is a service that translates domain names into IP addresses, helping devices communicate easily within a network environment.

3. **Dynamic Host Configuration Protocol (DHCP)**

DHCP automatically assigns IP addresses and network configurations to devices connected to the network, simplifying network management.

4. **Group Policy Object (GPO)**

A set of centralized rules and configurations used to manage user and computer settings within a Windows domain environment.

IP TABLE

Device	IP Address	Subnet Mask	Default Gateway	DNS Server	Role
Domain Controller (Server)	192.168.10.1	255.255.255.0	None (Directly connected)	192.168.10.1	AD DS, DNS, DHCP, GPO
DHCP Clients	192.168.10.10 – 192.168.10.100 (Range)	255.255.255.0	192.168.10.1	192.168.10.1	DHCP Client

Operating system virtual machine(VMWARE) specification:

1. Windows Server 2022
 - 2GB RAM
 - 60GB Storages
 - 1 CPU
 - 1 Internal Network Adapters (VMnet3)
2. Windows 2022 Client
 - 2GB RAM
 - 60GB Storages
 - 1 CPU
 - 1 Internal Network Adapters (VMnet3)

Do All Installations (Services)

On the Server

Open Server Manager

Manage > Add Roles and Features > Next Until Server Roles then Check DHCP DNS and AD DS > Next until Install

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Select server roles'. On the right, it says 'DESTINATION SERVER WIN-HFUNTPBHKTS.network1.com'. A left-hand navigation pane lists steps: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles' (highlighted in blue), 'Features', 'Confirmation', and 'Results'. The main area contains the instruction 'Select one or more roles to install on the selected server.' Below this is a list of roles with checkboxes. The roles listed are: Active Directory Certificate Services (unchecked), Active Directory Domain Services (Installed) (checked), Active Directory Federation Services (unchecked), Active Directory Lightweight Directory Services (unchecked), Active Directory Rights Management Services (unchecked), Device Health Attestation (unchecked), DHCP Server (Installed) (checked), DNS Server (Installed) (checked), Fax Server (unchecked), File and Storage Services (2 of 12 installed) (checked), Host Guardian Service (unchecked), Hyper-V (unchecked), Network Controller (unchecked), Network Policy and Access Services (unchecked), Print and Document Services (unchecked), Remote Access (unchecked), Remote Desktop Services (unchecked), Volume Activation Services (unchecked), Web Server (IIS) (unchecked), and Windows Deployment Services (unchecked). To the right of the list is a 'Description' section for 'Active Directory Certificate Services (AD CS)', stating it is used to create certification authorities and related role services. At the bottom, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

DESTINATION SERVER
WIN-HFUNTPBHKTS.network1.com

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.
☒ Active Directory Domain Services (Installed)	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☒ DHCP Server (Installed)	
☒ DNS Server (Installed)	
☐ Fax Server	
☒ File and Storage Services (2 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Controller	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	

< Previous Next > Install Cancel

DHCP Configuration

Installation & Authorization:

1. Server Manager > Manage > Add Roles and Features.
2. Check DHCP Server and click Install.
3. After installation, click the Flag icon > Complete DHCP configuration.
4. Click Commit to authorize the server in Active Directory.

Creating a Scope:

1. Open Tools > DHCP.
2. Expand your server > right-click IPv4 > New Scope.
3. Name: LAN_Scope.
4. Start IP: 192.168.10.10, End IP: 192.168.10.100.
5. Add Exclusions: (Optional) Skip if you don't need to reserve specific IPs within that range.
6. Configure DHCP Options: Select "Yes, I want to configure these options now".
 - o 006 DNS Servers: Ensure your server IP (192.168.10.1) is listed.
7. Click Activate Scope and Finish.

DHCP Configuration

DHCP Configuration

DHCP

File Action View Help

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DHCP	Start IP Address	End IP Address	Description
win-hfuntpbhks.network1.com	192.168.10.10	192.168.10.100	Address range for distribution

win-hfuntpbhks.network1.com

- IPv4
 - Server Options
 - Scope [192.168.10.0] Lan Scope
 - Address Pool
 - Address Leases
 - Reservations
 - Scope Options
 - Policies
 - Policies
 - Filters
 - IPv6

DHCP

File Action View Help

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DHCP	Client IP Address	Name	Lease Expiration	Type	U
win-hfuntpbhks.network1.com	192.168.10.10	CLIENT.network1.c...	5/1/2026 1:44:14 AM	DHCP	00

win-hfuntpbhks.network1.com

- IPv4
 - Server Options
 - Scope [192.168.10.0] Lan Scope
 - Address Pool
 - Address Leases
 - Reservations
 - Scope Options
 - Policies
 - Policies
 - Filters
 - IPv6

DNS Configuration

DNS translates names (like *google.com*) into IP addresses. It is automatically installed with AD DS, but needs two final touches.

A. Set Up Forwarders (For Internet Access)

1. Open Tools > DNS.
2. Right-click your Server Name > Properties > Forwarders tab.
3. Click Edit and type 8.8.8.8 (Google's DNS). This allows your local users to browse the internet.

B. Create a Reverse Lookup Zone

1. In DNS Manager, right-click Reverse Lookup Zones > New Zone.
2. Follow the wizard: Primary Zone > To all DNS servers in this domain > IPv4.
3. Network ID: Enter the first three parts of your IP: 192.168.10.
4. Click Finish. This allows the server to identify a computer's name just by its IP address.

DNS Configuration

DNS Configuration

DNS Manager

File Action View Help

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DNS		Name	Type	Data
▼	WIN-HFUNTPBHKTS	(same as parent folder)	Start of Authority (SOA)	[4], win-hfuntpbhks, hos...
▼	Forward Lookup Zones	(same as parent folder)	Name Server (NS)	win-hfuntpbhks.
>	_msdcs.network1.com	network1	Host (A)	192.168.10.1
>	com			
>	network1.com			
>	Reverse Lookup Zones			
>	Trust Points			
>	Conditional Forwarders			

DNS Manager

File Action View Help

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DNS		Name	Type	Data
▼	WIN-HFUNTPBHKTS	(same as parent folder)	Start of Authority (SOA)	[4], win-hfuntpbhks, hos...
▼	Forward Lookup Zones	(same as parent folder)	Name Server (NS)	win-hfuntpbhks.
>	_msdcs.network1.com	192.168.10.1	Pointer (PTR)	network.
>	com	192.168.10.10	Pointer (PTR)	CLIENT.network1.com.
>	network1.com			
▼	Reverse Lookup Zones			
>	10.168.192.in-addr.ar			
>	Trust Points			
>	Conditional Forwarders			

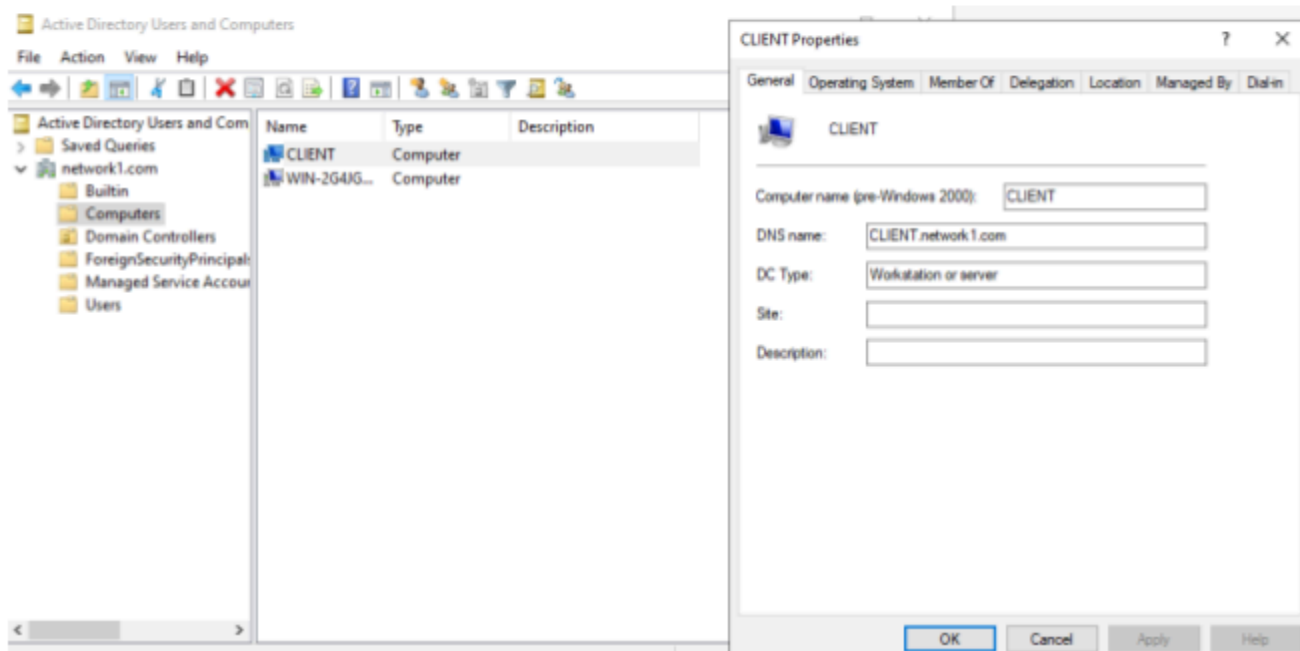
AD DS Configuration Server & Client

A. Installing the Role

1. In Server Manager, click Manage > Add Roles and Features.
2. Click Next until you reach Server Roles.
3. Check Active Directory Domain Services. In the pop-up, click Add Features.
4. Click Next through the rest of the wizard and click Install.

B. Promoting the Server to a Domain Controller

1. Once installation is finished, click the Notifications (Flag icon) in the top bar.
2. Select "Promote this server to a domain controller".
3. Choose "Add a new forest".
4. Root domain name: Type network1.com.
5. DSRM Password: Create a password (you will need this if the database ever crashes).
6. Click Next through the options (the system will handle DNS and NetBIOS).
7. Click Install. The server will automatically restart.



AD DS Configuration Server & Client

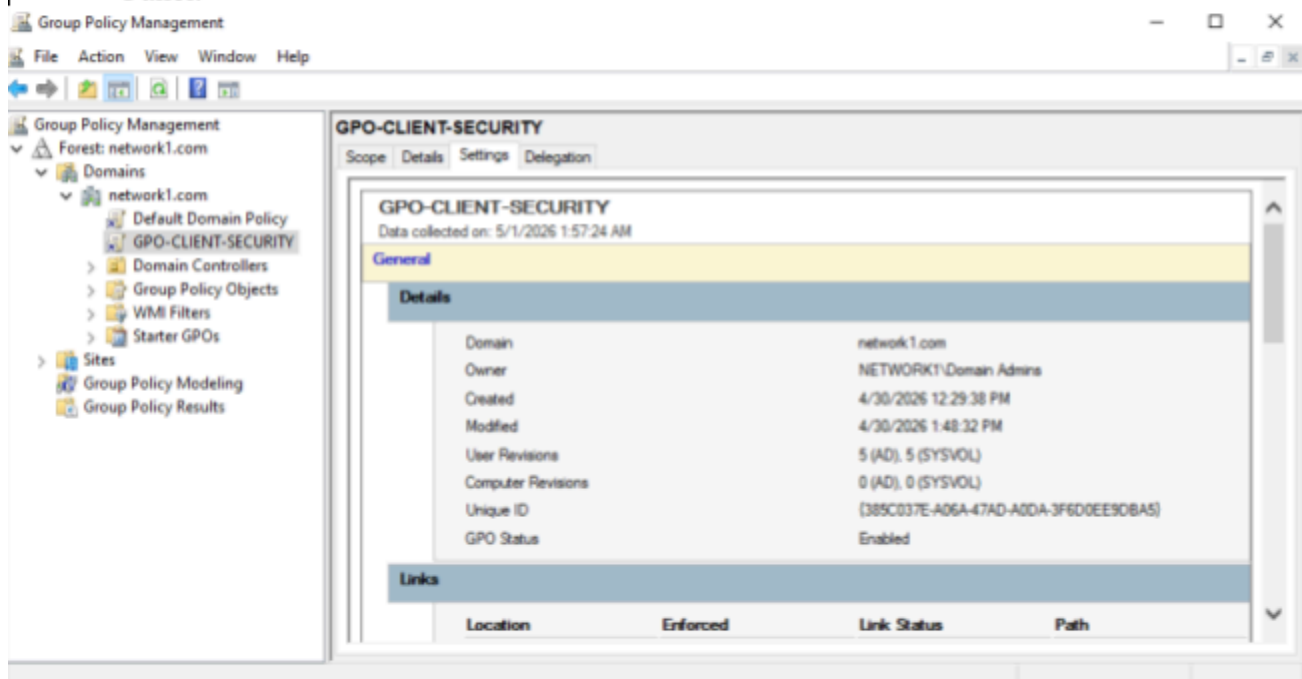
Connect the Client to the Domain

1. On the CLIENT computer, ensure the network settings are set to "Obtain an IP address automatically".
2. Open System Properties > Computer Name tab > Change.
3. Select Domain and type network1.com.
4. Enter the server's Administrator username and password.
5. Restart the client. Success Check:
 - On the Server, open Active Directory Users and Computers. The computer CLIENT will be visible in the Computers folder.
 - In the DHCP console, you will see 192.168.10.10 assigned to the client under Address Leases.

GPO Configuration

If you have installed AD DS, GPO is included in it
GPOs are rules that apply to everyone in the domain.

1. Open Tools > Group Policy Management.
2. Expand your forest > Domains > network1.com.
3. Right-click network1.com > Create a GPO in this domain, and Link it here....
4. Name: GPO-CLIENT-SECURITY.
5. Right-click your new GPO > Edit.
6. Example Policy: Go to *User Configuration > Policies > Administrative Templates > Control Panel*.
 - Enable "Prohibit access to Control Panel".
7. Now, any user who logs in to a computer in this domain will be unable to open the Control Panel.



Testing DHCP on the Client

In terminal run “ping 192.168.10.1” (Ip Server)

```
C:\Users\Administrator.NETWORK1>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

In terminal run “ping 192.168.10.10” (Ip Client)

```
C:\Users\Administrator.NETWORK1>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Testing DHCP on the Client

In terminal run “ipconfig /all”

```
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . . . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-CF-D4-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b980:529b:cb5c:477%5(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Friday, May 1, 2026 9:45:49 AM
    Lease Expires . . . . . : Wednesday, May 6, 2026 7:53:21 AM
    Default Gateway . . . . . : 
    DHCP Server . . . . . : 192.168.10.1
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-86-08-3B-00-0C-29-CF-D4-3A
    DNS Servers . . . . . : 192.168.10.1
                           192.168.23.2
    Primary WINS Server . . . . . : 192.168.23.2
    NetBIOS over Tcpip. . . . . : Enabled
```

Make sure DCHP Enabled : Yes and the IP matches what we set

Testing DNS on the Client

[network1.com](#) = 192.168.10.1

This is according to our DNS server settings

In terminal run “nslookup [network1.com](#)”

```
C:\Users\Administrator.NETWORK1>nslookup network1.com
Server: network
Address: 192.168.10.1

Name: network1.com
Addresses: 192.168.10.1
```

In terminal run “ping [network1.com](#)”

```
C:\Users\Administrator.NETWORK1>ping network1.com

Pinging network1.com [192.168.10.1] with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=4ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=3ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
```

Testing AD DS on the Client

Test AD DS (Active Directory)

Check the connection to the Domain Controller:

Check if the client can authenticate to AD:

In terminal run “nslookup 192.168.10.10”

```
C:\Users\Administrator.NETWORK1>nslookup 192.168.10.10
Server: network
Address: 192.168.10.1

Name: CLIENT.network1.com
Address: 192.168.10.10
```

Focus on CLIENT.network1.com

Then in terminal run “ping CLIENT network1.com”

```
C:\Users\Administrator.NETWORK1>ping CLIENT.network1.com

Pinging CLIENT.network1.com [fe80::286a:7fcb:32da:7b31%2] with 32 bytes of data:
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms

Ping statistics for fe80::286a:7fcb:32da:7b31%2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

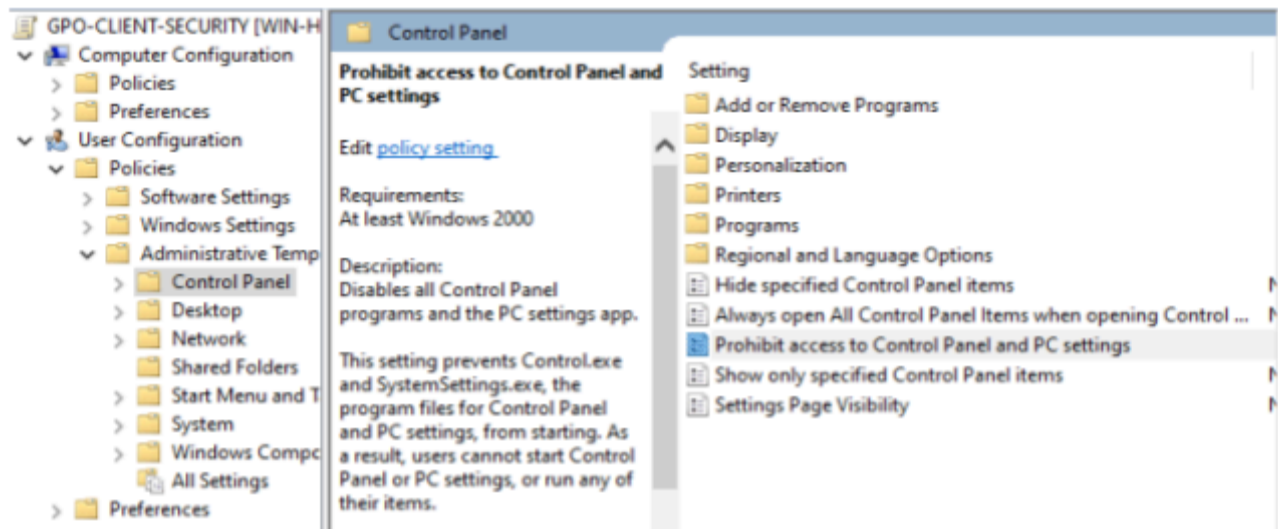
Testing GPO on the Server & Client

In this case, we will use an example, not providing access to the control panel on the client.

On the Server

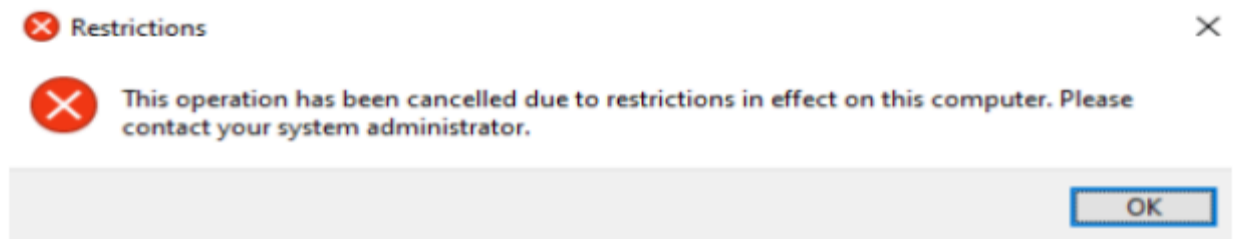
In terminal run “gpmc.msc”

1. User Configuration
2. Policies
3. Administrative Templates
4. Control Panel
5. Prohibit access to Control Panel and PC Settings (Click 2 times)
6. Enable then Click Ok



On the Client

Open Control Panel



If an error occurs, it means the GPO was successful.

This is the display when client tries to open control panel.

Requirements

Hardware :

1. Lenovo Thinkpad X1 Carbon

Operating System :

1. Windows Server 2022 64-bit
2. Windows Client 2022 64-bit

Software :

1. Vmware
2. Google.docs
3. Google Chrome

Project File Details :

<u>No</u>	<u>Files</u>	<u>Remarks</u>
1.	2CS1 Project 3.pdf	Content of the paper
2.	Project 3 Kel 3.pdf	Presentation file